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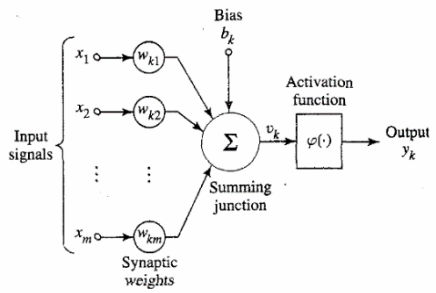
Department of Artificial Intelligence and Machine Learning

	Test -1 –Scheme and Solution	**Max. Marks: 50+10**
Semester: V	**UG**	**Duration: 1 Hrs + ½ Hr**
Course Title: Artificial Neural Networks and Deep Learning		**Course Code: AI253IA**

Part A

Sl No	Questions	M	BT	CO
1.	A perceptron has weights . For input , compute the net input and output (step activation). 0.5 and 1	1	3	1
2.	Compute the output of a sigmoid neuron if . **0.6900**	1	2	1
3.	The learning rate (η) is a positive scalar that scales weight updates during training (how big a step we take in parameter space). If η is too large training may diverge or oscillate; if too small convergence is very	1	2	2
4.	0.5,1.05, 2.10	2	4	2
5.	Can learn **nonlinearly separable** functions (hidden layers let them combine features to solve problems single-layer perceptrons cannot, e.g., XOR).Can **approximate complex mappings** (universal approximation for continuous functions) and learn hierarchical features.	2	3	1
6.	11	2	3	1
7.	Sequential (online): update weights after each training example — faster immediate updates, introduces noise that can help escape local minima. Batch: accumulate errors/gradients over the whole training set and update weights once per epoch — smoother, more stable gradient estimate but needs more memory and slower per-epoch updates.	1	2	2

Part B

Sl No	Questions	M	BT	CO
1a.	 - 2M, Explanation -2M	**04**	**2**	**1**
1b.	**Bias (b) Net Input Sigmoid Output Observation**	**06**	**3**	**1**

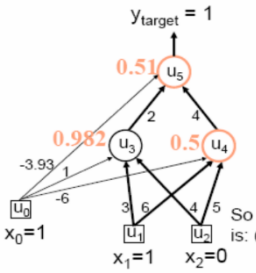
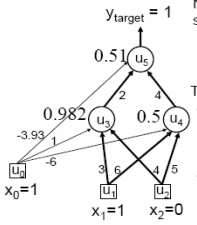
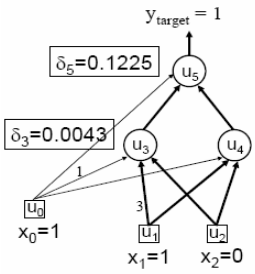
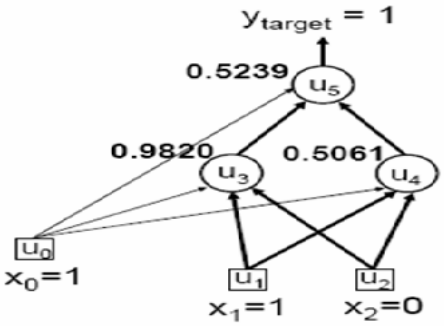
	0.1 0.7 0.668 Normal activation -3M 0.5 1.1 0.750 Higher bias → higher output-3M						
2a.						05	3	2

Learning Rule	Basic Principle	Weight Update Formula (Conceptual)	Type of Learning	Key Features	Practical Example
Hebbian Learning	"Neurons that fire together, wire together." — strengthens connections between simultaneously active neurons.	$\Delta w = \eta \cdot x \cdot y$ (where x = input, y = output)	**Unsupervised**	Reinforces correlations between inputs and outputs; no error or teacher signal.	**Associative memory** (e.g., pattern storage in Hopfield networks).
Competitive Learning	Neurons compete; only one (winner) neuron updates its weights toward the input (Winner-Takes-All).	Only winning neuron's weights updated: $\Delta w = \eta \cdot (x - w)$	**Unsupervised**	Promotes specialization — neurons become tuned to different input patterns or clusters.	**Clustering / Self-Organizing Maps (Kohonen networks)**
Error-Correction Learning	Adjusts weights to minimize the difference (error) between desired and actual outputs.	$\Delta w = \eta \cdot (d - y) \cdot x$	**Supervised**	Uses a teacher signal; aims to reduce prediction error; basis for backpropagation.	**Perceptron / MLP training for classification or regression**

2b				05	4	2

Quantity	Expression	Value
Output (y)	(w^T x)	−0.7
Error (e)	(d - y)	3.7
Updated Weights	(w + η e x)	[0.685, −0.23]

3	Explain the Perceptron Learning Algorithm with its update rule and convergence theorem. Why does the perceptron fail for non-linearly separable problems?			

4	 **Unit** **activation** **output** a_j y_j u_3 4.00 0.982 u_4 0.00 0.500 u_5 0.04 **0.510** **(network output)** So the error for this training example is: $(y_{target} - y_5)$=(1 - 0.510)= 0.490  Now compute delta values starting at the output: δ_5 = $y_5(1 - y_5)(y_{target} - y_5)$ = 0.51(1 - 0.51) x 0.49 = **0.1225** Then for hidden units: δ_4 = $y_4(1 - y_4)\, w_{45}\, \delta_5$ = 0.5(1 - 0.5) x 4 x 0.1225 = **0.1225** δ_3 = $y_3(1 - y_3)\, w_{35}\, \delta_5$ = 0.982(1-0.982) x 2 x 0.1225 = **0.0043**  ◆ Set learning rate η = 0.1 Change weights by: $\Delta w_{ij} = \eta \delta_j y_i$ ◆ e.g.bias weight on u_3: Δw_{03} = $\eta \delta_3 x_0$ =0.1*0.0043*1 = 0.0004 So, new w_{03} ⌫ w_{03}(old)+Δw_{03} =1+0.0004=1.0004 ◆ and likewise: w_{13} ⌫ 3 + 0.0004 = 3.0004 i j w_{ij} δ_j y_i Updated w_{ij} 0 3 **1** 0.0043 1.0 **1.0004** 1 3 **3** 0.0043 1.0 **3.0004** 2 3 **4** 0.0043 0.0 **4.0000** 0 4 **-6** 0.1225 1.0 **-5.9878** 1 4 **6** 0.1225 1.0 **6.0123** 2 4 **5** 0.1225 0.0 **5.0000** 0 5 **-3.92** 0.1225 1.0 **-3.9078** 3 5 **2** 0.1225 0.9820 **2.0120** 4 5 **4** 0.1225 0.5 **4.0061**  On next forward pass: The new activations are: y_3 = f (4.0008) = 0.9820 y_4 = f (0.0245) = 0.5061 y_5 = f (0.0955) = **0.5239** Thus the new error $(y_{target} - y_5)$=(1 - 0.5239)=0.476 has been reduced by **0.014** (from **0.490** to **0.476**)	**14**	**4**	**3**
5		**10**	**4**	**2**

5) ①

a). $= (0.1 \times 1) + (-0.3 \times 10) + (0.2 \times 12) + (-0.2 \times 13)$

$= 0.1 + (-3) + (2.4) + (-2.6)$

$= -3.1$

net input.

sigmoid output $y = f(\text{net}^{i/p})$

$y = f(-3.1)$ Activation function used is sigmoid

$$\boxed{y = \frac{1}{1+e^{-3.1}} = 0.0431}$$

$\therefore y < 0.5 \Rightarrow$ misclassified. (not bus/ Car)

b). $(t-y) = 1 - 0.0431 = 0.9569$

$y(1-y) = 0.0431\ (1 - 0.0431) = 0.0413$

$\eta (t-y)\, y(1-y) = 0.1 \times 0.9569 \times 0.0413 = 0.00394$

$\Delta W_i = \eta (t-y) \cdot y(1-y) x_i$ Considering $\eta (t-y) \cdot y(1-y) = k$

$= k \cdot x_i$

$\Delta W_0 = k \cdot x_0 = 0.00394\ (1) = 0.00395$

$\Delta W_1 = k \cdot x_1 = 0.00394\ (10) = 0.03947$

$\Delta W_2 = k \cdot x_2 = 0.00394\ (12) = 0.04737$

$\Delta W_3 = k \cdot x_3 = 0.00394\ (13) = 0.05131$

iii) Determine the new weights after the update:

$w_i = w_i + \Delta w_i$

$w_0 = 0.1 + 0.00395 = 0.10395 \Rightarrow \approx 0.104$

$w_1 = -0.3 + 0.03947 = -0.260 \quad \approx -0.261$

$w_2 = 0.2 + 0.04737 = 0.247 \quad \approx 0.247$

$w_3 = -0.2 + 0.05131 = -0.14869 \quad \approx -0.14/0.15$

(iv) $net = w_0 x_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$

$= 0.1039(1) + (-0.261 \times 10) + (0.247 \times 12) + (-0.15 \times 13)$

$= 0.1039 - 2.61 + 2.964 - 1.95$

$= -1.4921$

$y = \frac{1}{1+e^{-1.49}} = 0.1881 \approx 0.20$ again $y < 0.5$

misclassified - so it is Cat.

v) $(1, 5, 7, 7) = -1.4$

$\sigma(-1.4) = 0.1981$ - Correctly classifies as Cat.

so for new weights,

$net_{new} = 0.103(1) + (-0.260)(5) + (0.247(7)) + (-0.14)(7) = -0.508$

$\sigma(-0.508) = 0.376 < 0.5$

$\therefore$ output has increased from 0.198 to 0.376. → Cat

M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks CIE	16	30	14	-	-	-	13	16	31	-	-